

Layer	Technology	Version	Deployment Option	Server Requirement	License
OS (App Server)	Ubuntu Server 22.04 LTS / RHEL 9.3+ / Windows Server 2022 Standard / Datacenter (minimum) and above	LTS	Installed directly	8 Core CPU, 32 GB RAM, 500 GB NVMe SSD (RAID 1)(minimum)	Free / Paid
OS (DB Server)	RHEL 9.3+ (or) Windows Server 2022 Datacenter / Standard (minimum) and above	LTS	Installed directly	12 Core CPU, 64 GB RAM, 1 TB NVMe SSD (minimum)	Paid
Python	Python	3.11.x	Direct install OR inside Docker	App Server	Open source
Django	Django	5.2.x (or 4.2 LTS)	Direct install OR Docker container	App Server	Open source
Database	MySQL Enterprise	8.1	Direct install (recommended)	Dedicated DB Server	Oracle Enterprise
Web Server	Nginx	1.24+	Direct install OR Docker container	App Server	Open source
WSGI	Gunicorn	Latest stable	System service OR inside Docker	App Server	Open source
Cache	Redis	7.x	Direct install OR Docker	App Server	Open source

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Queue	Celery + Redis	5.x	System service OR Docker	App Server	Open source
Container Platform (Optional)	Docker Engine	24.x LTS	Recommended for app layer	App Server	Open source/paid
Container Orchestration (Optional)	Docker Compose	2.x	Optional	App Server	Open source/paid
Monitoring	Prometheus + Grafana	2.x / 10.x	Separate server	6 Core, 32 GB RAM, 1 TB SSD	Open source
Logging	ELK Stack (Elasticsearch, Logstash, Kibana)	8.x	Separate server	6 Core, 32 GB RAM, 1–2 TB SSD	Open source
Backup	RAID 10 + Cloud Backup	-	Automated	Daily + Offsite encrypted	-

Above table is for Server Specification

Developer Environment – Hardware Requirements

1. **CPU:** (Intel i5/i7 13th+ or AMD Ryzen 5/7)
2. **RAM:** Minimum 16 GB, recommended 32 GB
3. **Storage:** 512 GB SSD minimum, recommended 1 TB NVMe SSD
4. **GPU:** needed for AI/ML modules or computer vision libraries like DeepFace and also for loading and training AI models. Any modern GPU (VRAM 8GB-minimum, 16GB-recommended and above)

Developer Environment – Software Requirements

1. **Operating System:** Windows 11 Pro
2. **Python:** Version 3.11.x
3. **Django:** Version 5.2.
4. **Database:** MySQL 8.1 Enterprise/community or PostgreSQL 15.
5. **Web Server:** Nginx (optional locally) to simulate production environment.
6. **WSGI Server:** Gunicorn, run locally to test production-like behavior.
7. **Cache / Queue:** Redis 7.x and Celery 5.x, optional via Docker for local development.
8. **Container Platform (Recommended for Production):** Docker Enterprise Edition (Docker Engine 24.x and Docker Compose 2.x) – Paid license required for government/enterprise deployment to ensure support, security, and compliance. Used to run containers for Django, Redis, Celery, and MySQL (PAID)
9. **IDE / Code Editor:** VS Code (Free version)
10. **Version Control:** Git, latest stable, installed locally.
11. **Browser:** Chrome or Firefox (latest) for frontend testing.
12. **Optional Tools:** VirtualBox or Vagrant for simulating multi-server development environments if needed.